When **M** acts on this eigenvector, it multiplies the vector by a different number, namely −1. On the other hand, if **M** acts on the vector

$$\begin{pmatrix} 1 \\ 0 \end{pmatrix},$$

the vector is *not* simply multiplied by a number. **M** alters the direction of the vector as well as its magnitude.

Just as the vectors that get multiplied by numbers when **M** acts on them are called eigenvectors of **M**, the constants that multiply them are called *eigenvalues*. In general, the eigenvalues are complex numbers. Here is an example that you can work out for yourself. Take the matrix

$$\mathbf{M} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

and show that the vector

$$\begin{pmatrix} 1 \\ i \end{pmatrix}$$

is an eigenvector with eigenvalue $-i$.

Linear operators can also act on bra-vectors. The notation for multiplying $<B|$ by **M** is

$$\langle B|\mathbf{M}.$$

I will keep the discussion short by telling you the rule for this type of multiplication. It is most simple in component form. Remember that bra-vectors are represented in component form as row vectors. For example, the bra $<B|$ might be represented by